

Module 7

Financial Management

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer 1(a)

Using the CAPM, the cost of common stock:

$$\begin{aligned}K_e &= R_f + [R_m - R_f] \times \text{beta} \\&= 0.03 + (0.12 - 0.03) \times 0.85 \\&= 10.65\%\end{aligned}$$

The cost of preferred stock:

$$\begin{aligned}K_p &= \text{HK\$}100 \times 5\% / \text{HK\$}96 \\&= 5.21\%\end{aligned}$$

The pre-tax cost of debt:

$$\begin{aligned}K_d &= \text{yield to maturity} \\&= 6\%\end{aligned}$$

Answer 1(b)

Market value of common stock = HK\$13 x 10 million = HK\$130 million

Market value of preferred stock = HK\$96 x 1 million = HK\$96 million

Market value of debt (bonds) = HK\$1,000 x 120,000 = HK\$120 million

Total market value = HK\$(130 + 96 + 120) million = HK\$346 million

$$W_E = E/V = \text{HK\$}130 \text{ million} / \text{HK\$}346 \text{ million} = 37.6\%$$

$$W_P = P/V = \text{HK\$}96 \text{ million} / \text{HK\$}346 \text{ million} = 27.7\%$$

$$W_D = D/V = \text{HK\$}120 \text{ million} / \text{HK\$}346 \text{ million} = 34.7\%$$

$$\text{WACC} = 10.65\% \times 37.6\% + 5.21\% \times 27.7\% + 6\% \times (1 - 16.5\%) \times 34.7\% = 7.19\%$$

Answer 1(c)

The Efficient Market Hypothesis ("EMH") states that all known information is reflected in stock prices immediately upon being known to the public. Any new information becoming available will be quickly reflected in stock market prices. As such, stock prices accurately reflect all available information at any given time. This implies that it is impossible to beat the market because prices are efficient.

There are three forms of market efficiency:

Weak Form

The weak form EMH states that stock prices are reflections of all past stock information including all past price and volume trading information.

Under this form of EMH, future prices cannot be forecasted by studying past price behaviour. It means that technical analysis is not reliable for making investment decisions. In other words, stock prices follow a random series of movements.

Semi-strong Form

The semi-strong form EMH states that stock prices reflect all publicly known and available information in addition to the weak form set of information. The public information is those relevant to the stock in question, including earnings, dividends, corporate investments and management changes etc.

Semi-strong form efficient markets will have all past stock information and publicly available information reflected in current prices. As such, it is not possible to analyse publicly available information by fundamental analysis in order to identify stocks that are mispriced.

Strong Form

Strong form EMH states that stock prices reflect all public and private information including both the weak form and semi-strong form sets of information. All stock prices are fully efficient so it is not possible for investors to beat the market by analysing past price behaviour, publicly available information or insider information.

Answer 1(d)(i)

In a semi-strong market efficiency, current share prices reflect all relevant information which is publicly available. However, if the information has not been released to the public, the share prices will not be affected.

If the restructuring plan is only approved by the board of directors and has not been announced to the public yet, this will be regarded as insider information within the board of directors of Kimberlite.

As such, the market will not respond to the private information and there will be no change in the share prices of Kimberlite.

Answer 1(d)(ii)

In a semi-strong market efficiency, public information will be reflected very quickly in the share prices.

If the restructuring plan has been announced to the public, this means it would become public information and share prices of Kimberlite would be adjusted very quickly.

Since the restructuring plan will have a positive impact on the profitability of Kimberlite, the share prices should increase accordingly right after the announcement.

Answer 2(a)

Required rate of return = 3% + 2% = 5%

Proposal A

Annual net revenue = HK\$4 x 12,000 – HK\$2,000 = HK\$46,000

$NPV_A = -HK\$100,000 + HK\$46,000 \times (1 - (1 + 0.05)^{-5}) / 0.05 = HK\$99,156$

Proposal B

Annual net revenue = (HK\$4 – HK\$2.2) x 12,000 = HK\$21,600

$NPV_B = HK\$21,600 \times (1 - (1 + 0.05)^{-5}) / 0.05 = HK\$93,517$

Proposal A should be accepted as it has a higher NPV.

Answer 2(b)

Accounting rate of return for Proposal B

Total revenue for the five-year period = HK\$2.2 x 12,000 x 5 = HK\$132,000

Total depreciation and maintenance expenses for the five-year period = HK\$70,000 + HK\$1,500 x 5 = HK\$77,500

Total accounting return for 5 years = HK\$132,000 – HK\$77,500 = HK\$54,500

Average accounting return = HK\$54,500 / 5 = HK\$10,900

Average assets = (HK\$70,000 + HK\$0) / 2 = HK\$35,000

Accounting rate of return = HK\$10,900 / HK\$35,000 = 31.14%

Since the accounting rate of return is much higher than the required rate of return of 12%, Proposal B is viable for Hematite.

Answer 2(c)

The objective of the power supply company to launch the SEPS has likely to do with corporate social responsibility ("CSR"). CSR is a business model that seeks to combine financial profits with social and environmental benefits in the business strategy. Companies will minimise the negative impacts that their operations have on society, the environment, the economy etc. that are not directly related to their business.

Even though SEPS does not make any financial sense does not mean the power supply company should not undertake it. The project will have positive impact to the environment as solar energy is a renewable energy which would not pollute the environment compared with other non-renewable resources such as fossil fuels. The SEPS will also encourage community participation in the development of renewable energy to address the climate crisis.

For the power supply company itself, the SEPS strengthens its CSR practice and will also have intangible benefits for the company such as strengthening moral and employee loyalty, enhancing the goodwill and reputation of the company. It considers outcomes based on long term sustainability and not just short term returns to shareholders.

Answer 3(a)(i)

Dividend growth model:

$$\begin{aligned}\text{Total share value} &= D_0 (1 + g) / (r - g) \\ &= (\text{HK\$}15 \text{ million} \times 0.35 \times 1.1) / (0.13 - 0.1) = \text{HK\$}192.5 \text{ million}\end{aligned}$$

$$\text{Value of each share} = \text{HK\$}192.5 \text{ million} / 50 \text{ million} = \text{HK\$}3.85$$

$$\text{Number of shares to be issued} = \text{HK\$}20 \text{ million} / \text{HK\$}3.85 = 5.19 \text{ million shares}$$

$$\text{Percentage of ownership} = 5.19 / (5.19 + 50) = 9.40\%$$

Answer 3(a)(ii)

P/E ratio method:

The total value of share = Earnings x P/E ratio

Since Andesite is not a listed company, the P/E should be lower than similar listed companies.

A discount of 20% is suggested to adjust the P/E ratio for valuation purpose.

$$= \text{HK\$}15 \text{ million} \times 14 \times (1 - 20\%) = \text{HK\$}168 \text{ million}$$

$$\text{Value of each share} = \text{HK\$}168 \text{ million} / 50 \text{ million} = \text{HK\$}3.36$$

$$\text{Number of shares to be issued} = \text{HK\$}20 \text{ million} / \text{HK\$}3.36 = 5.95 \text{ million shares}$$

$$\text{Percentage of ownership} = 5.95 / (5.95 + 50) = 10.63\%$$

Answer 3(b)

Venture capitalists seek to provide funding in return for the shares in private companies with high growth potential. For entrepreneurs who need start-up capital but find difficulties in accessing traditional sources of funding, venture capitalists may provide the necessary funding.

Since venture capitalists are running a high-risk business model, they will use a number of tactics to mitigate their risk.

The probable tactics employed by venture capitalists are as follows:

- They will provide funding in stages.
- They may require personal guarantees from owners/ entrepreneurs.
- They may create syndicates to share the risk if the funding amount is too high for them.
- They will not invest in the industry if they have little knowledge about it. They are normally specialised in one or two industries only.
- They may require the right to appoint representatives to the company's board of directors.

Venture capitalists typically require equity interest issued by the target company for their funding. The equity can be in the form of common stock, or preferred stock with the option to convert into common stock in the future. They generally will hold the investment for a period of three to seven years until it is successful.

Venture capital agreements may include exit provisions such as the expected timing, method and price etc. In order to achieve a high return, venture capitalists will normally realise their investment through a stock market listing.

Answer 3(c)

The advantages of obtaining funding from venture capitalists:

Venture capitalists usually require an equity stake in target companies. Andesite can avoid to make any payments to the venture capitalists if the company does not have sufficient cash flows at the early stage of its expansion. If the financing is through a bank loan, the company needs to pay interest regardless of whether it can generate sufficient cash flow to do so.

Since equity financing can lower the debt-to-equity ratio, it will make future debt financing (e.g., bonds issue or bank loans) easier.

Vivianite should have maintained a valuable business network that Andesite can tap into. Vivianite can introduce potential customers, suppliers and joint venture partners etc. to Andesite.

Since Vivianite is specialised in the Andesite's industry, experienced leadership and advice are available from the asset management team of Vivianite.

The disadvantages of obtaining funding from venture capitalists:

By appointing representatives to the board of Andesite, Vivianite will have a significant influence over the board policy. However, the interest of Vivianite may not be in line with Andesite.

Bank loan has a cheaper cost than issuing equity to Vivianite especially when interest is a tax deductible expense.

Shareholding will be diluted. Profit needs to be shared with Vivianite if Andesite becomes very profitable in future.

Venture capital agreement may include restrictive covenants that may reduce the flexibility of managing the business of Andesite.

Answer 4(a)(i)

Days' Inventory = average inventory / cost of goods sold x 365
[HK\$(1,230,000 + 1,600,000) / 2] / HK\$5,400,000 x 365 = 96 days

Days' Receivables = average receivables / net sales x 365
[HK\$(1,280,000 + 1,580,000) / 2] / HK\$8,050,000 x 365 = 65 days

Days' Payables = average payables / cost of goods sold x 365
[HK\$(1,250,000 + 1,290,000) / 2] / HK\$5,400,000 x 365 = 86 days

Answer 4(a)(ii)

The Cash Conversion Cycles ("CCC") for the industry average
= Days' Inventory + Days' receivables – Days' Payables
= 65 + 55 – 60 = 60 days
CCC for Zoisite = 96 + 65 – 86 = 75 days

Answer 4(a)(iii)

The CCC for Zoisite is 75 days, while comparing with the industry average of 60 days, it seems that Zoisite is not doing well in terms of working capital management.

In general, if Zoisite wants to improve the CCC, it may try to reduce inventories, tighten the credit control for customers or delay payments to suppliers.

For reducing the inventories, it seems that Zoisite should have some room to improve as the inventory days of Zoisite is 96 days while the industry average is 65 days. Zoisite appears to hold inventories one month longer than the industry average. The disadvantage of holding inventory is that it is a major working capital investment which ties up funds that may be used in the other areas of the business. Zoisite should formally review its inventory management techniques to see if there are any areas for improvement. It may consider implementing the just-in-time system or improve inventory ordering procedures etc.

For tightening credit control, it seems that Zoisite should also have some room to improve as the receivables days of Zoisite is 65 days while the industry average is 55 days. At present, customers will pay only 65 days after the sale date. Zoisite should investigate whether the credit terms are too long or is it simply that the customers do not pay when the payments are due. If the credit terms are too long, Zoisite should try to tighten the credit terms without losing revenue. If it is due to delaying payments by customers, Zoisite should review its collection procedures so that payments can be received on time.

For delaying payments, it seems that it is not applicable to Zoisite's situation as the payables days of Zoisite is 86 days while the industry average is 60 days. Zoisite appears to be performing much better than the industry average. However, Zoisite should avoid the disadvantage of losing reliable suppliers if the delaying of payments is not within the agreed terms of the purchase contracts. Zoisite may also consider taking early settlement discounts if they can be justified by comparing the cost of financing working capital.

Answer 4(b)

Net working capital is defined as the current assets less the current liabilities.

Working capital management refers to the day-to-day management of the company's current assets and liabilities.

Working capital management is important because many businesses may be forced to go bankrupt due to the inability to meet short-term payment obligations even though the businesses are making a positive accounting profit.

One of the objectives of managing working capital is to ensure the liquidity of the company. The company should have enough money to pay bills when they fall due. Otherwise, the company may be forced into compulsory liquidation to repay the outstanding liabilities.

The other objective is to make sure that any spare cash should earn adequate interest by investing it as profitably as possible. Funds tied up in working capital may not earn the required return for the investors. Companies with high working capital may have a negative impact on the return on capital employed.

* * * END OF EXAMINATION PAPER * * *